

Title

Cloud-Based Student Learning Management System (LMS)

Project Overview

Build a cloud-hosted LMS where students can register, log in, view courses, download learning materials, submit assignments, and view announcements. Admins manage students, upload content, and post announcements.

Business Problem

An EdTech startup needs a scalable online learning platform without maintaining physical servers. AWS provides scalable, reliable infrastructure.

Objectives

Deploy the web app on AWS EC2, store files in S3, implement authentication with MySQL/PostgreSQL, monitor using CloudWatch, and manage code with GitHub.

Student Portal

Registration/Login, View Courses, Download Learning Materials, Submit Assignments.

Admin Portal

Upload Course Content, Post Announcements, Manage Students.

Architecture

EC2 hosts the application; S3 stores PDFs/videos/assignments; MySQL stores users, courses and submissions; CloudWatch monitors health and logs.

Technologies

HTML, CSS, JavaScript, Python Flask or Node.js, MySQL, AWS EC2, AWS S3, CloudWatch, GitHub.

Implementation Phases

1. Create AWS account & EC2. 2. Develop LMS. 3. Connect database. 4. Configure S3. 5. Configure CloudWatch. 6. Deploy and test.

Deliverables

Running LMS website, architecture diagram, source code, deployment documentation, monitoring dashboard, final presentation.